

Conditionals, Couplings and Toric Geometry: Kernels, the Coupling Nerve, Cumulants and a Fukaya Programme

Revised draft 3

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October 6, 2026

Abstract

We organize the conditional and coupling structure of the finite distribution monad D into four layers and separate what is proved from what is conjectural.

- (1) *The base.* Markov kernels are the 1-cells of the Kleisli category, and couplings are 2-cells whose vertical composition is the gluing lemma. The natural whiskerings are not functorial, so the couplings do not form a 2-category. Their image under the cost map does: a strict, graded, Lawvere-enriched 2-category.
- (2) *The fibre.* The base-relative posterior tower is a functor on the Kleisli category. Its category of elements is a discrete opfibration over it, and every grade of the shadow bounds the distance between transported contexts at every level of the tower.
- (3) *The coupling nerve.* Couplings of all arities form the simplicial object $D(Y^{\bullet+1})$, with marginalization as faces. Every 2-horn fills. The fillers form a product of coupling polytopes, and the maximum-entropy filler is the one whose third face is the vertical composite. For $n \geq 3$ the faces of an n -horn form a cyclic cover, so filling can fail. The failure is measured by the contextual fraction, and $D(Y^{\bullet+1})$ is not a Kan complex, although $Y^{\bullet+1}$ is.
- (4) *Linearization.* Centred couplings are second mixed cumulants, and whiskering defects are the Boolean cumulants of whiskering viewed as a noncommutative expectation. Into the square-zero cone of the circulation ideal, whiskering is an A_∞ -functor with exactly two components. Its cohomology keeps only the marginals, and the coupling data are homotopical.

On top of these layers sits the geometry. Coupling polytopes of nondegenerate marginals are Delzant, so they are moment polytopes of toric manifolds whose Kähler class is the product coupling and whose canonical metric restricts to the Fisher metric. Their walls and chambers, monotone cases, Floer theory and Bridgeland stability conditions are treated as before. Conjectures are stated with correct types, and a companion paper [4] computes every construction explicitly.

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*Verification scripts are listed in [section 11](#); the worked examples are in the companion [4].

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1 Introduction

Two structures sit on top of the finite distribution monad D . The base-relative posterior tower [2] fixes an outcome base B and solves a *strict integrability* problem: it carries a mixed distributive law satisfying all four Beck axioms. The conditional/coupling structure [1] lets the base vary along Markov kernels and compares kernels by couplings: it solves a *transport* problem. This paper assembles them, together with a simplicial and a homotopical layer, into one framework.

Layer	Objects and structure	Status
I. Base	finite sets, kernels, coupling 2-cells; the graded shadow	strict 2-category on grades only (section 2)
II. Fibre	the tower over each base; transport along kernels	discrete opfibration; grades act on fibres (section 3)
III. Nerve	$D(Y^{\bullet+1})$; horns, fillers, boundary fibres	2-horns fill; higher horns obstructed by contextuality (section 4)
IV. Linearization	circulations, cumulants, cones	whiskering is a two-component A_∞ -functor into a cone (section 5)
Geometry	coupling polytopes as moment polytopes; stability	proved in the nondegenerate case (sections 6 to 8)
Programme	families over chambers; Fukaya-type functor	conjectural (section 9)

The layers interlock. A 2-cell is a 1-simplex of the nerve, and vertical composition is the canonical filler of an inner 2-horn. The tangent spaces of the filler polytopes are circulations, and those are where the cumulants of layer IV live. The tower of layer II is transported along the 1-cells of layer I, and the grades of layer I bound the transport. The polytopes of layer III are, in the nondegenerate case, moment polytopes, and that is where the geometry enters. What does *not*

hold is equally important. The couplings themselves do not form a 2-category, the nerve is not Kan, and the toric picture stops at boundary fibres of 3-way tables.

2 Layer I: kernels, couplings and the graded shadow

$\mathbf{D}$ is the finitely supported distribution monad on $\mathbf{Set}$ and $\mathbf{Kl}(\mathbf{D})$ its Kleisli category. A 1-cell $k: B \rightsquigarrow B'$ is a Markov kernel $k: B \rightarrow \mathbf{D}(B')$, acting on states by $k^\#(p)(b') = \sum_b p(b)k(b)(b')$. The results of this section are proved in [1, §7]; we restate them for reference.

Proposition 2.1 (1-cells [1, Props. 7.2, 7.3]). *$k^\#$ is affine, $(h \circ k)^\# = h^\# k^\#$, $k^\# \circ \mu = \mu \circ \mathbf{D}(k^\#)$, and $\mathrm{TV}(k^\# p, k^\# q) \leq \mathrm{TV}(p, q)$. Attribution sets are transported along k , and precision can only be lost.*

Definition 2.2 (Coupling 2-cells). For 1-cells $f, g: X \rightsquigarrow Y$, a coupling 2-cell $\alpha: f \Rightarrow g$ is a kernel $\alpha: X \rightarrow \mathbf{D}(Y \times Y)$ with marginals $f(x)$ and $g(x)$. Its cost is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$; the identity is the diagonal coupling; the vertical composite glues along the common marginal, $(\beta \cdot \alpha)(x)(y, y'') = \sum_{y'} \alpha(x)(y, y')\beta(x)(y', y'')/g(x)(y')$.

Theorem 2.3 (Witnesses and shadow [1, Thms. 7.6, 7.8]).

- (i) *For fixed X, Y , 1-cells and coupling 2-cells form a category under gluing, strictly associative and unital. Cost is subadditive and $\min_\alpha c(\alpha) = d(f, g) = \max_x \mathrm{TV}(f(x), g(x))$.*
- (ii) *Whiskering by a deterministic precomposition is strictly functorial. Whiskering by a stochastic precomposition (mixing couplings) or by a postcomposition (copying on equal pairs, independent on unequal pairs) preserves identities, marginals and the cost bound, but is not functorial.*
- (iii) *Graded 2-cells $f \Rightarrow_\varepsilon g$, $\varepsilon \geq d(f, g)$, with grades adding, form a strict 2-category on kernel 1-cells, and d makes $\mathbf{Kl}(\mathbf{D})$ a Lawvere-enriched category.*

Remark 2.4 (Witness layer and shadow). The cost map $\alpha \mapsto c(\alpha)$ sends the witness layer onto the graded shadow. It preserves identities and sends composites to grades no larger than the sum. The shadow is an image of the witness layer, not a substructure of it, and it is the only strict 2-category in sight: with couplings as 2-cells, (ii) rules out a 2-category or a bicategory with these whiskerings. The polytopes $\mathrm{Cpl}(f(x), g(x))$ are the spaces of 2-cells, not further cells; their geometry is the subject of [sections 4 and 6](#).

3 Layer II: the tower as a fibration

The base-relative tower over B has context algebra $A_B = \mathbf{D}B \times \mathbf{D}^2 B \times \mathbf{D}^3 B \times \mathrm{Sh}(B)$, reader comonad $\mathbf{C}^B(Y) = Y \times A_B$ and strict law λ^B [2, §9]. With $\mathrm{Hyp} = \Theta \times \mathbf{D}(-)$, a kernel $k: B \rightarrow \mathbf{D}(B')$ acts on hypotheses by $\mathrm{Hyp}(k)(\theta, \rho) = (\theta, k^\# \rho)$, and we put $A_k = k^\# \times \mathbf{D}(k^\#) \times \mathbf{D}^2(k^\#) \times \mathbf{D}(\mathrm{Hyp}k)$.

Theorem 3.1 (The tower is a functor on the Kleisli category [2, Prop. 9.14, Cor. 9.15, Prop. 9.16]).

- (i) *A_k is a morphism of $\mathbf{D}$ -algebras, preserves coherent contexts, commutes with the canonical section σ , and $A_{h \circ k} = A_h A_k$; for deterministic k it is the earlier base change.*
- (ii) *$\mathrm{id} \times A_k$ is a comonad morphism $\mathbf{C}^B \Rightarrow \mathbf{C}^{B'}$ commuting with the laws $\lambda^B, \lambda^{B'}$.*

- (iii) *Admissibility data, coherent lifts and the effective sets are transported; certificates pull back along $f' \mapsto k \triangleright f'$; failure levels can only rise and the secondary obstruction can only fall.*
- (iv) *Conditioning on an outcome observed after the channel equals conditioning on the pulled-back predicate $k \triangleright 1_{e'}$, then changing base.*

Definition 3.2 (The total category). Let $\mathcal{E}$ be the category whose objects are pairs (B, z) with B a finite set and $z \in A_B^{\text{coh}}$ a coherent context, and whose morphisms $(B, z) \rightarrow (B', z')$ are kernels k with $A_k(z) = z'$. Let $\pi: \mathcal{E} \rightarrow \text{Kl}(\mathcal{D})$ forget z .

Proposition 3.3 (Split opfibration). π is a discrete opfibration, and in particular a split opfibration. The cocartesian lift of $k: B \rightarrow B'$ at (B, z) is $k: (B, z) \rightarrow (B', A_k z)$. Through $\text{id} \times A_k$, the comonads $\mathcal{C}^B$ and the laws λ^B are carried along the lifts.

Proof. $\mathcal{E}$ is the category of elements of the functor $B \mapsto A_B^{\text{coh}}$, $k \mapsto A_k$, which is a functor by [theorem 3.1\(i\)](#); categories of elements of set-valued functors are discrete opfibrations. The statement about comonads and laws is [theorem 3.1\(ii\)](#). $\square$

Proposition 3.4 (Grades act on fibres). Let $f, g: B \rightarrow \mathcal{D}(B')$ and let $z = (\rho, \Pi, \Omega, \Sigma)$ be a coherent context. Put TV on $\mathcal{D}B'$, the Kantorovich metric W_1 over TV on $\mathcal{D}^2 B'$, the Kantorovich metric over that on $\mathcal{D}^3 B'$, and on hypotheses the metric that is TV of predictives for equal descriptors and 1 otherwise. Then every component of $A_f(z)$ and $A_g(z)$ is within $d(f, g)$. So a graded 2-cell $f \Rightarrow_\varepsilon g$ gives ε -close transported contexts at every level.

Proof. $\text{TV}(f^\# \rho, g^\# \rho) \leq \sum_b \rho(b) \text{TV}(f(b), g(b)) \leq d(f, g)$ by convexity. For Π , push Π forward along $\rho \mapsto (f^\# \rho, g^\# \rho)$. This is a coupling of $\mathcal{D}(f^\#) \Pi$ and $\mathcal{D}(g^\#) \Pi$ of cost $\mathbb{E}_\Pi \text{TV}(f^\# \rho, g^\# \rho) \leq d(f, g)$. Level three and the shadow are the same argument one level up. The companion verifies the bound in a worked example, where all three levels give $\frac{2}{15} \leq \frac{1}{6}$ [[4](#), §11]. $\square$

The proof of [theorem 3.1](#) uses one fact beyond naturality: Kleisli extension is a morphism of free $\mathcal{D}$ -algebras. All other components of A_k are pushforwards along functions. So the strict integrability of each fibre and the transport between fibres are compatible.

4 Layer III: the coupling nerve

Definition 4.1 (The coupling nerve). For a finite set Y , let $N_k(Y) = \mathcal{D}(Y^{k+1})$, with face maps d_i that marginalize out the i th coordinate and degeneracies s_i that push forward along the diagonal repeating it. Since $Y^{\bullet+1}$ is a simplicial set (the Čech nerve of $Y \rightarrow 1$) and $\mathcal{D}$ is a functor, $N(Y) = \mathcal{D}(Y^{\bullet+1})$ is a simplicial set. The relative version over a set X has k -simplices the kernels $X \rightarrow \mathcal{D}(Y_0 \times \cdots \times Y_k)$, with all structure taken pointwise in x .

A 1-simplex of $N(Y)$ is a coupling of its two vertices. A coupling 2-cell $\alpha: f \Rightarrow g$ between kernels $X \rightsquigarrow Y$ is an X -indexed 1-simplex with vertices f and g . A horn Λ_k^n in N is a compatible family of laws on the n faces other than the k th, and a filler is an n -simplex with those faces. In probabilistic terms a horn is an empirical model on the cover of $\{0, \dots, n\}$ by those faces, and a filler is a global section [[17](#)].

Theorem 4.2 (2-horns fill, and composition is the canonical filler). Let p_{01} and p_{12} be compatible laws on $Y_0 \times Y_1$ and $Y_1 \times Y_2$ with common marginal p_1 , the data of an inner horn Λ_1^2 .

- (i) The fillers form the product, over x_1 with $p_1(x_1) > 0$, of the transportation polytopes of the slices $p(\cdot, x_1, \cdot)$ with marginals $p_{01}(\cdot, x_1)$ and $p_{12}(x_1, \cdot)$. If every slice has full support and nondegenerate marginals, the filler polytope is Delzant.
- (ii) The gluing filler $p^{\text{gl}} = p_{01} p_{12} / p_1$ is the unique filler of maximal entropy, and its face d_1 is the vertical composite of the two couplings.
- (iii) Any filler differs from p^{gl} , slice by slice, by $p_1(x_1) \text{Cov}(1_{\{x_0\}}, 1_{\{x_2\}} \mid x_1)$, a circulation of the slice.

The outer horns Λ_0^2 and Λ_2^2 fill in the same way, by gluing over the shared vertex.

Proof. (i) Each marginal constraint involves one value of x_1 , so the constraints decouple into slices. Each slice is a two-way transportation polytope, Delzant under the stated conditions (theorem 6.4), and a product of Delzant polytopes is Delzant with respect to the product lattice. (ii) $\log p^{\text{gl}}$ is a sum of a function of (x_0, x_1) and a function of (x_1, x_2) , so it is orthogonal to every tangent vector of the filler polytope. So p^{gl} is a critical point of the strictly concave entropy on the polytope, hence its unique maximizer. Equivalently, $H(X_0 X_1 X_2) \leq H(X_0 X_1) + H(X_1 X_2) - H(X_1)$, with equality exactly under conditional independence. Summing p^{gl} over x_1 gives the gluing formula of definition 2.2. (iii) is a direct computation. $\square$

So the 2-cells of layer I are the 1-simplices of the nerve, and their vertical composition is the canonical filling of inner 2-horns. The other fillers are conditional cumulants, which is the link to layer IV.

Theorem 4.3 (Higher horns). *Let $n \geq 3$.*

- (i) The n faces of Λ_k^n , viewed as subsets of $\{0, \dots, n\}$, form a cover that is not acyclic. For $n = 2$ it is acyclic.
- (ii) Hence, for $|Y| \geq 2$, there are compatible horn data in $N(Y)$ with no filler.
- (iii) For compatible horn data, the minimal unmodeled weight with the deterministic global assignments as sources, each face weighted $1/n$, equals the contextual fraction of the horn, and it vanishes exactly when the horn fills. For the parity horns on two outcomes (uniform on assignments of fixed parity on each face, with exactly one face odd) it is $\frac{1}{2}$ for every 3-horn and $\frac{3}{4}$ for every 4-horn.

Proof. (i) Use Graham's reduction for acyclic hypergraphs [18]: repeatedly delete vertices in a single edge and edges contained in others; the hypergraph is acyclic if and only if this empties it. In Λ_k^n the vertex k lies in all n faces and every other vertex lies in $n - 1$ of them. No face contains another, so for $n \geq 3$ no reduction step applies. For $n = 2$ the two faces share one vertex and reduce away. (ii) By Vorob'ev's theorem [25], every compatible family extends to a global law exactly on acyclic covers. Data on two outcomes embed into any Y with $|Y| \geq 2$. (iii) A filler is a global section, so this is [1, Thm. 7.12] applied to the horn's cover. The parity values are computed with exactly verified certificates in [4, §10]. $\square$

Corollary 4.4 (The nerve is not Kan). *For $|Y| \geq 2$, $N(Y) = D(Y^{\bullet+1})$ fills every 2-horn but not every n -horn for $n \geq 3$. In particular it is not a Kan complex, although $Y^{\bullet+1}$ is. The monad D does not preserve horn filling, because it does not turn compatible families over cyclic covers into global laws, and the obstruction is the contextual fraction.*

Proposition 4.5 (Boundary fibres). *Prescribing all three faces of a 2-simplex, that is all three 2-margins of a 3-way table on $Y_0 \times Y_1 \times Y_2$, gives a planar 3-way transportation polytope of dimension $(n_0 - 1)(n_1 - 1)(n_2 - 1)$ when it is nonempty and generic. It can be empty even for compatible faces, because the cover by the three edges of a triangle is cyclic.*

- (i) *For $n_0 = 2$ it is a capacitated two-way transportation polytope, whose constraint matrix is totally unimodular, so it is Delzant whenever it is simple.*
- (ii) *Every rational polytope is a planar $r \times s \times 3$ transportation polytope [19], so boundary fibres are not toric in general. Already for $3 \times 3 \times 3$, generic instances are simple but not smooth: they are moment polytopes of toric orbifolds [22], not of manifolds [4, §10].*

Proof. For $n_0 = 2$ the second layer is $m - y$, where y is the first layer, m is the $(1, 2)$ -margin, and y has fixed row and column sums with $0 \leq y \leq m$. A bipartite incidence matrix stacked with an identity is totally unimodular, so edges are ± 1 cycle vectors and the argument of theorem 6.4 applies at simple vertices. (ii) combines the cited theorem with the computation in the companion. $\square$

So the toric regime of the nerve is exactly the 1-simplices and the 2-horns. Boundary fillings leave it, and higher horns can be empty.

5 Layer IV: cumulants and the square-zero cone

Proposition 5.1 (Couplings are second cumulants). *For $\alpha \in \text{Cpl}(f, g)$, $\alpha - f \otimes g$ is the cross-covariance $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$ of the indicator vectors, a circulation. The product coupling is the point where all mixed cumulants vanish, and it is the only point of rank one, the Birch point of the independence model [23]. For level-two couplings of second-order states,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \text{Cov}_\Pi(\rho, \rho'),$$

so flattening keeps only the second mixed cumulant of the two random distributions.

Fix $f \in \mathcal{D}(Y)$ and let A_f be the matrices on $Y \times Y$ whose row and column sums are multiples of f . With the gluing product $a \cdot b = a \text{diag}(f)^{-1} b$ and unit $\text{diag}(f)$, A_f is an associative unital algebra, spanned by the couplings of f with itself. The circulations I_f form a two-sided ideal, and $A_f/I_f \cong \mathbb{R}$ records the total mass. Whiskering by a kernel h is a linear unital map $h_*: A_f \rightarrow A_{h\#f}$, which need not be multiplicative: a noncommutative expectation in the sense of homotopy probability theory [20, 21]. Its Boolean cumulants [24] are $b_1 = h_*$, $b_2(a, b) = h_*(a \cdot b) - h_*a \cdot h_*b$, and so on over interval partitions. The whiskering defect of theorem 2.3(ii) is $-b_2$.

Theorem 5.2 (Whiskering is a two-component A_∞ -functor). *Let $C = A_{h\#f} \oplus \varepsilon I_{h\#f}$, with ε of degree -1 , $\varepsilon^2 = 0$ and $d(\varepsilon x) = x$. Then C is a differential graded algebra with $H^0(C) = A_{h\#f}/I_{h\#f} \cong \mathbb{R}$ and $H^{-1}(C) = 0$: it is the mapping cone of $I \hookrightarrow A$. The maps*

$$F_1 = h_*, \quad F_2 = \varepsilon b_2, \quad F_n = 0 \quad (n \geq 3)$$

form an A_∞ -functor $A_f \rightarrow C$.

Proof. Every b_2 is a circulation, because both composites couple the same marginals, so F_2 lands in εI . C is the trivial extension of A by the bimodule εI , and d is a derivation because I is an ideal. Arity two of the functor equation is $dF_2 = b_2$, true by definition. In arity three, with $F_3 = 0$, it reads $b_2(ab, c) - b_2(a, bc) = h_*a \cdot b_2(b, c) - b_2(a, b) \cdot h_*c$. Both sides equal $h_*a \cdot h_*(bc) - h_*(ab) \cdot h_*c$, which says that b_2 is the Hochschild coboundary of h_* . In arity $n \geq 4$ every term involves some F_k with $k \geq 3$ or a product $F_2 \cdot F_2 \in \varepsilon^2 I = 0$. $\square$

Remark 5.3. This settles, in the minimal model, the question of the previous draft whether whiskering is an A_∞ -functor. The higher Boolean cumulants are nonzero [4, §9], but they are not needed here. They appear as higher components only in larger models such as the free cone, as in the bar-construction description of cumulants. The defects span all of I for three and four points [4, §10], so H^0 of the cone generated by the defects alone is also A/I .

The four layers together. Under the cone, all coupling information is homotopical: H^0 keeps only masses, so the 1-cells of layer I, while the couplings live in degree -1 , in the circulations. These are the tangent spaces of the polytopes of layer III. The cumulants of [proposition 5.1](#), the defects of [theorem 5.2](#) and the deviations of 2-horn fillers from the canonical filler ([theorem 4.2\(iii\)](#)) are all circulations, and the Fisher metric of [proposition 6.5](#) measures them. Layer II carries the tower along the same kernels, with the grades of layer I as bounds.

6 Coupling polytopes are moment polytopes

6.1 The polytope and its lattice

Let $f \in D(Y)$ and $g \in D(Y')$ have full support, with $|Y| = n$, $|Y'| = m$. (For coupling 2-cells $Y = Y'$; rectangular polytopes arise when gluing contexts [1, Prop. 7.11].) The *coupling polytope* is

$$\text{Cpl}(f, g) = \left\{ \alpha \in \mathbb{R}_{\geq 0}^{Y \times Y'} : \sum_{y'} \alpha(y, y') = f(y), \sum_y \alpha(y, y') = g(y') \right\}.$$

Its tangent space is the space of *circulations*, matrices with zero row and column sums. The integer circulations form a lattice Λ of rank $(n-1)(m-1)$; a basis is $E_{ij} - E_{i,m} - E_{n,j} + E_{n,m}$ for $i < n$, $j < m$.

Proposition 6.1 (Interior and Fisher metric). *Cpl(f, g) has dimension $(n-1)(m-1)$, its relative interior is the set of strictly positive couplings, and on it the Fisher metric $\mathcal{I}_\alpha(v, w) = \sum v(y, y')w(y, y')/\alpha(y, y')$ is positive definite on circulations.*

Proof. The product coupling $f \otimes g$ is strictly positive, so no inequality $\alpha \geq 0$ is an implicit equality, and the relative interior of the polytope in its affine hull is $\{\alpha > 0\}$. There $\mathcal{I}_\alpha(v, v) = \sum v^2/\alpha > 0$ for $v \neq 0$. $\square$

Remark 6.2. The first draft argued nondegeneracy through a radical and then distinguished “the Fisher metric” from “the Fisher–Rao metric”. These are the same metric, the Fisher information metric. The square-root embedding is a way of computing its geodesic distance [1, Prop. 8.1].

6.2 Nondegeneracy implies Delzant

Definition 6.3 (Walls). The pair (f, g) is *degenerate* if $\sum_{y \in I} f(y) = \sum_{y' \in J} g(y')$ for some nonempty proper subsets $I \subsetneq Y$ and $J \subsetneq Y'$. These equations define hyperplanes, the *walls*, in the space of pairs of marginals; the *chambers* are the connected components of the complement of the walls.

Theorem 6.4 (Coupling polytopes are Delzant). *If (f, g) is nondegenerate, then $\text{Cpl}(f, g)$ is a Delzant polytope with respect to Λ : it is simple, its facet normals are rational, and at each vertex the primitive edge vectors form a basis of Λ . Its edges at a vertex are the fundamental cycles of a spanning tree.*

Proof. Regard α as a weighting of the edges of the complete bipartite graph on $Y \sqcup Y'$. *Vertices are forests.* If the support of a feasible α contains a cycle, adding $\pm\epsilon$ alternately around it stays feasible, so α is not a vertex. If the support is a forest, the marginal equations determine α on it uniquely (peel off leaves), so α is a vertex. *Nondegeneracy makes them spanning trees.* Every node has positive marginal and hence an incident edge of the support. If the support forest had two or more components, take one with row set I and column set J . Then $\sum_I f = \sum_J g$; I and J are nonempty, and if $I = Y$ then $\sum_J g = 1$ forces $J = Y'$ and the component is everything. So I and J are proper, contradicting nondegeneracy. *Simplicity and smoothness.* At a vertex with spanning tree T , the zero cells are the $nm - (n + m - 1) = (n - 1)(m - 1)$ non-tree cells c . Each closes a unique cycle in $T \cup \{c\}$, giving a circulation C_c with entries in $\{0, \pm 1\}$ and $C_c(c) = 1$. A circulation w equals $\sum_{c \notin T} w(c)C_c$, because the difference is a circulation supported on the tree, hence zero. So the feasible directions at the vertex form the cone $\{\sum_c t_c C_c : t \geq 0\}$ on exactly $\dim \text{Cpl}(f, g)$ independent generators (simplicity), and the C_c are integral and form a $\mathbb{Z}$ -basis of Λ , since integer coefficients $w(c)$ suffice (smoothness). The facet normals are the coordinate functionals $\alpha \mapsto \alpha(c)$, which are rational. $\square$

By Delzant's theorem [9] there is a compact connected toric symplectic manifold $(X_{f,g}, \omega)$ of real dimension $2(n - 1)(m - 1)$, unique up to equivariant symplectomorphism, whose moment map $\mu: X_{f,g} \rightarrow \text{Cpl}(f, g)$ has image the coupling polytope. The fibre over an interior coupling is a Lagrangian torus $T^{(n-1)(m-1)}$, and the fibre over a vertex is a point.

Proposition 6.5 (Fisher metric and Guillemin's potential). *Suppose every inequality $\alpha(c) \geq 0$ defines a facet of $\text{Cpl}(f, g)$. The canonical Kähler metric of $X_{f,g}$ constructed by Guillemin [13] has, in action-angle coordinates over the interior, the form $\text{Hess } G$ in the polytope directions, with symplectic potential*

$$G(\alpha) = \frac{1}{2} \sum_c \alpha(c) \log \alpha(c).$$

Hence the Fisher metric on $\text{Cpl}(f, g)$ is twice the restriction of the canonical Kähler metric to the real section, and the product coupling $f \otimes g$, the coupling of maximal entropy, is the unique critical point of G on the polytope.

Proof. Guillemin's potential is $\frac{1}{2} \sum_i \ell_i \log \ell_i$ over the facets, where $\ell_i \geq 0$ are the facet inequalities written with primitive inward normals. Here $\ell_c(\alpha) = \alpha(c)$, and the coordinate functional is primitive on Λ because some integer circulation has entry 1 at c . Then $\text{Hess } G(v, v) = \frac{1}{2} \sum_c v(c)^2 / \alpha(c)$. The critical-point condition is that $\log \alpha$ is orthogonal to all circulations, that is $\log \alpha(y, y') = r(y) + s(y')$, which together with the marginals gives $\alpha = f \otimes g$. Uniqueness follows from strict convexity. $\square$

If some cell is not a facet, for instance when $f(y) + g(y') > 1$ forces $\alpha(y, y') > 0$, the potential omits that term and the identity holds only up to those terms.

6.3 Walls, chambers and monotone cases

Proposition 6.6 (Chambers). *The set of spanning trees that support vertices, and hence the combinatorial type of $\text{Cpl}(f, g)$ and the toric manifold $X_{f,g}$ up to equivariant diffeomorphism, is constant on each chamber. The type can change across a wall. For $f = (\frac{1}{2}, \frac{1}{2})$ and $g = (t, \frac{2}{3} - t, \frac{1}{3})$ the polytope is a hexagon for $t < \frac{1}{2}$ and a quadrilateral for $t > \frac{1}{2}$, both Delzant: two facets become redundant across the wall $g_1 = f_1$.*

Proof. The tree solution on a spanning tree T assigns to each tree edge the value $\sum_I f - \sum_J g$ (up to sign) for the bipartition (I, J) obtained by deleting that edge. So feasibility of T depends

only on the signs of such differences, which are constant on chambers. The example is checked in `check_conditionals.py`. $\square$

Proposition 6.7 (Monotone cases). *Assume (f, g) is nondegenerate and every cell defines a facet. Then $X_{f,g}$ is monotone if and only if f and g are uniform, and uniform marginals are nondegenerate exactly when $\gcd(n, m) = 1$. In particular, in the square case $Y = Y'$ of coupling 2-cells no nondegenerate pair is monotone. On the wall, the uniform 2×2 pair still gives a Delzant segment, $X = \mathbb{CP}^1$, with the product coupling as its midpoint; the uniform $n \times n$ pair with $n \geq 3$ gives a polytope that is not simple.*

Proof. For a Delzant polytope written with primitive normals, the toric manifold is monotone exactly when some interior point has equal lattice distance to all facets: the classes of the toric divisors span H^2 subject to the linear relations given by the normals, c_1 is the sum of the divisor classes, and $[\omega]$ is the sum weighted by the facet distances. Here the distances are the entries $\alpha(c)$, so they are equal exactly when α is constant, which forces f and g to be uniform; then $\alpha = f \otimes g$. Uniform marginals are nondegenerate iff $a/n \neq b/m$ for $0 < a < n$, $0 < b < m$, iff $\gcd(n, m) = 1$, which fails when $n = m$. The two wall cases are checked in `check_conditionals.py` (T3). $\square$

Example 6.8 (The degree-6 del Pezzo surface). For f uniform on two points and g uniform on three, $\text{Cpl}(f, g)$ is a hexagon. In the basis of Λ above, its facet normals are $\pm e_1$, $\pm e_2$ and $\pm(e_1 + e_2)$. This is the fan of $\mathbb{CP}^2$ blown up at three points, the del Pezzo surface of degree 6. All six entries of the product coupling equal $\frac{1}{6}$, so it is the equidistant point. The case 3×4 (uniform) gives a Delzant polytope of dimension 6 with 96 vertices and all 12 cells facets, again monotone.

7 The imported layer: Floer theory of the special fibre

We now quote results from symplectic topology. They are not proved or checked here.

Nondisplaceability. For a monotone toric manifold, the torus fibre over the point at equal distance from all facets (the *special fibre*) is a stem in the sense of Entov and Polterovich, hence nondisplaceable [10]. For toric Fano manifolds, Cho and Oh show that a torus fibre with a flat $U(1)$ connection has nonvanishing Floer cohomology when it corresponds to a critical point of the disc potential [8]; see [7] for the Clifford torus and [11] for general compact toric manifolds.

Consequence for couplings. In the monotone cases of [proposition 6.7](#), the special fibre lies over the product coupling $f \otimes g$. For [example 6.8](#), the disc potential of the monotone special fibre is, up to the factor $T^{1/6}$, the Laurent polynomial

$$W(x, y) = x + y + \frac{1}{x} + \frac{1}{y} + xy + \frac{1}{xy},$$

read off from the facet normals, and it has a critical point at $(x, y) = (1, 1)$ (checked exactly), so the special fibre with trivial holonomy has nonzero Floer cohomology by [8].

The maximal coupling, by contrast, has zero entries whenever $f \neq g$, so it lies on the boundary of $\text{Cpl}(f, g)$, where the torus fibre collapses (`check_conditionals.py`, T6). The first draft's conjecture that the critical points are the maximal couplings should therefore be replaced. The geometrically distinguished coupling is the product coupling, which is also the maximum-entropy coupling and the critical point of Guillemin's potential ([proposition 6.5](#)).

8 Stability conditions: a question

This subsection replaces a draft that proposed a Bridgeland stability condition whose central charge is the disc potential, whose heart is the committed states, and whose Harder–Narasimhan filtration is the obstruction tower. None of those identifications is well typed. A central charge is an additive map $K_0 \rightarrow \mathbb{C}$ and W is a function on a torus. Committed states are vertices of a polytope and form no abelian category. The obstruction tower assigns numbers to a claim; it does not filter an object. What can be said precisely is the following.

Bridgeland stability. A stability condition on a triangulated category $\mathcal{D}$ is a pair $(Z, \mathcal{A})$. Here $\mathcal{A}$ is the heart of a bounded t -structure, and $Z: K_0(\mathcal{D}) \rightarrow \mathbb{C}$ is additive, sends nonzero objects of $\mathcal{A}$ to the upper half-plane together with $\mathbb{R}_{<0}$, and admits Harder–Narasimhan filtrations; the support property is also required [6]. On any smooth projective surface X , and for any ample class ω , there is a stability condition with central charge

$$Z_\omega(E) = - \int_X e^{-i\omega} \text{ch}(E)$$

whose heart is a tilt of $\text{Coh } X$ [3, 15]. (A B -field can be added; we set it to zero.)

Proposition 8.1 (Marginals give stability conditions on the 2×3 surfaces). *Let (f, g) be nondegenerate in the 2×3 family, let X_Σ be the toric surface of its chamber, and let D_c be the toric divisor of the facet $\{\alpha_c = 0\}$. Put*

$$[\omega_{f,g}] = \sum_{c \text{ facet}} f_i g_j [D_c], \quad c = (i, j),$$

the product coupling read as divisor coefficients. Then:

- (i) $[\omega_{f,g}]$ is the class of the toric symplectic form with moment polygon $\text{Cpl}(f, g)$. It is independent of the base point used to write the polygon, and it is ample.
- (ii) $\omega_{f,g} \cdot D_c$ is the lattice length of the edge $\{\alpha_c = 0\}$ of the coupling polygon, and $\frac{1}{2}\omega_{f,g}^2$ is its lattice area.
- (iii) There is a stability condition $\sigma_{f,g} = (Z_{\omega_{f,g}}, \mathcal{A}_{\omega_{f,g}})$ on $D^b(\text{Coh } X_\Sigma)$. It depends continuously on (f, g) within the chamber, and

$$Z(\mathcal{O}_X) = \text{area } \text{Cpl}(f, g), \quad Z(\mathcal{O}_x) = -1, \quad Z(\mathcal{O}_{D_c}) = \frac{1}{2}D_c^2 + i \text{ length}\{\alpha_c = 0\}.$$

- (iv) As (f, g) approaches a wall from inside a chamber, one edge length tends to 0, its divisor E is a (-1) -curve, and $Z(\mathcal{O}_E) \rightarrow -\frac{1}{2}$, a point on the negative real axis.

Proof. (i) If a Delzant polygon is $\{u : \langle u, v_c \rangle + \lambda_c \geq 0\}$, the symplectic class is $\sum_c \lambda_c [D_c]$, up to the normalization of ω . Here the facet functions are $\alpha_c(u) = \langle u - u_0, v_c \rangle + \alpha_c(u_0)$ for any interior base point u_0 , and choosing $u_0 = f \otimes g$ gives the formula. Another choice of base point changes the coefficients by $\langle w, v_c \rangle$, which is a linear relation among the $[D_c]$. A Kähler class on a projective toric surface is ample. (ii) These are standard toric identities: the degree of ω on D_c is the lattice length of the corresponding edge, and the symplectic volume is the lattice area (Duistermaat–Heckman). (iii) Apply [3, 15] with $\omega = \omega_{f,g}$, which varies continuously with (f, g) . The values follow from $\text{ch}(\mathcal{O}_X) = 1$, $\text{ch}(\mathcal{O}_x) = [\text{pt}]$ and $\text{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$. (iv) By [proposition 6.6](#) and `check_stability_layer.py` (S5), the vanishing edge is that of a (-1) -curve. All identities are checked in all 18 chambers. $\square$

So the probabilistic data enter Bridgeland stability through the Kähler class, and nowhere else. The product coupling fixes the class, edge lengths of the coupling polygon are imaginary parts of central charges, and its area is $Z(\mathcal{O}_X)$. At the uniform point $\omega = -\frac{1}{6}K_X$ on the del Pezzo surface, with area $\frac{1}{12}$ and six edges of length $\frac{1}{6}$.

Questions.

- (Q1) *Across walls.* Crossing a wall blows up or blows down a point (proposition 6.6), and Orlov's theorem gives $D^b(\mathrm{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X) \rangle$ [16]. As the wall is approached, $Z(\mathcal{O}_E)$ tends to the negative real axis. Does the path $\sigma_{f,g}$ extend across the wall to a path through stability conditions glued along Orlov's decomposition? If so, the chamber structure of the marginals becomes a wall structure in a single space of stability conditions.
- (Q2) *Objects for couplings.* Is there a natural object $E_\alpha \in D^b(\mathrm{Coh} X)$ attached to an interior coupling α ? A candidate is the SYZ or mirror transform of the torus fibre over α with a flat connection. Does $Z(E_\alpha)$ have a probabilistic meaning, for instance in terms of the Fisher metric of proposition 6.5?
- (Q3) *Quantum corrections.* Iritani's integral structure identifies quantum-corrected central charges of objects of $D^b(\mathrm{Coh} X)$ with oscillatory integrals $\int e^{-W/z}$ of the mirror potential over integral cycles, and their asymptotics involve the $\widehat{\Gamma}$ -class [14, 12]. This is the precise sense in which the disc potential is related to central charges: through oscillatory integrals, not through its values. Do the corrected central charges, with $\omega = \omega_{f,g}$, define stability conditions on these surfaces? Does the monotone point $f \otimes g$ uniform play a special role?
- (Q4) *Higher dimensions.* For $(n-1)(m-1) \geq 3$ the toric manifolds have complex dimension at least three. There, the existence of stability conditions is known only in special cases, through Bogomolov–Gieseker type inequalities [5]. The construction of proposition 8.1 therefore extends only conditionally.

What is not claimed. We make no identification of the disc potential with a central charge, of committed states with a heart, of the obstruction tower with Harder–Narasimhan filtrations, or of hallucination with instability. The obstruction tower and the stability conditions above live on different objects (claims and admissibility data on one side, coherent sheaves on $X_{f,g}$ on the other), and any relation between them would have to be constructed.

9 The conjectural layer

Proposition 9.1 (The 2×3 family [4, §3]). *In the 2×3 family there are 18 chambers. Every toric surface $X_{f,g}$ is the degree-6 del Pezzo surface or a toric blow-down of it, and crossing one wall is exactly one toric blow-up or blow-down.*

Conjecture 9.2 (Family over the chamber complex). *For general $n \times m$, the toric manifolds $X_{f,g}$ form a family over each chamber, and crossing a wall is a toric birational transformation (in higher dimension possibly a flip) determined by the pair (I, J) defining the wall. On a wall, $X_{f,g}$ degenerates to a singular toric variety whose moment polytope is the non-simple $\mathrm{Cpl}(f, g)$.*

Question 9.3 (Location of Floer-nontrivial fibres). Outside the monotone cases, the fibres with nonvanishing Floer cohomology (possibly after bulk deformation) are located by the potential of [11]. Is the product coupling always among them, and do they have a statistical meaning?

Question 9.4 (Beyond the minimal cone). [Theorem 5.2](#) makes whiskering an A_∞ -functor into the square-zero cone, with two components. (i) In the free cone, do the higher Boolean cumulants give the higher components? (ii) Does the construction extend to the multi-object setting, an A_∞ -functor between linear categories whose objects are 1-cells, compatible with the nerve of [section 4](#)?

Conjecture 9.5 (Fukaya-type functor). *There is a functor from a category built from the Fukaya categories of the $X_{f,g}$ (with special fibres as distinguished objects) to a linearization of the coupling 2-cells, compatible with gluing on one side and composition of Lagrangian correspondences on the other.*

10 Next steps

The companion [\[4\]](#) computes every layer explicitly: the 2×3 family, the square 2×2 case, levels two and three, disc potentials, stability, cumulants, the nerve, the tower fibration and the family over the marginals. The open problems are:

- (1) *Across a wall.* Decide (Q1) of [section 8](#) for the wall $f_1 = g_1$ between the pentagon and hexagon chambers.
- (2) *General wall-crossing.* Prove [conjecture 9.2](#) beyond 2×3 .
- (3) *Higher horns.* Give explicit unfillable n -horns and their contextual fractions for all $n \geq 5$, and compute the best approximate fillers.
- (4) *Multi-object cone.* Answer [question 9.4\(ii\)](#), linking the cone to the nerve.
- (5) *Fukaya.* [Conjecture 9.5](#).

11 Computational checks

All checks pass.

- `check_glue.py` (30 checks): [proposition 2.1](#) and [theorem 2.3](#).
- `check_stochastic_basechange.py` (13 checks): [theorem 3.1](#); the hexagon of [example 6.8](#); the Delzant property for random nondegenerate 3×3 marginals; non-simplicity of the uniform 3×3 case; the Hessian identity and critical point of [proposition 6.5](#).
- `check_conditionals.py` (7 checks): [theorem 6.4](#), [propositions 6.6](#) and [6.7](#), and [example 6.8](#).
- `check_stability_layer.py` (6 checks): [proposition 8.1](#) in all 18 chambers of the 2×3 family.
- In the companion [\[4\]](#):
 - `check_tower_fibration.py` (7 checks): [theorem 3.1](#) and [proposition 3.4](#);
 - `check_nerve.py` (10 checks): [theorems 4.2](#), [4.3](#) and [5.2](#) and [proposition 4.5](#);
 - `check_cumulants.py` (8 checks): [proposition 5.1](#);
 - `check_polytope_companion.py`, `check_family.py` and `check_family_geometry.py`: [proposition 9.1](#).

A Corrections to the first draft

1. *Candidate disc potential.* $W_\beta(f) = \sum_g \int_{\text{Cpl}(f,g)} e^{-\beta c(\alpha)} d\nu$ sums over a continuum of 1-cells g and is not defined as written. Its Laplace limit $\min_{g,\alpha} c(\alpha)$ is always 0, attained at $g = f$, $\alpha = \Delta_f$, so it carries no information. The toric disc potential of $X_{f,g}$ is the correct object (section 7).
2. *Symplectic structure.* A symplectic form is antisymmetric and the Fisher metric is symmetric, so the restriction of the one cannot be the other. The correct statement is proposition 6.5: the Fisher metric is twice the canonical Kähler metric on the real section of $X_{f,g}$.
3. *Critical points.* “Critical points of $\pi: E \rightarrow \text{Kl}(\mathbb{D})$ ” is undefined, since π is a functor, not a smooth map. The replacement is the special fibre over the product coupling in the monotone cases, and the walls of definition 6.3 as the locus where the family degenerates.
4. *Monodromy and Dehn twists.* The family $(f, g) \mapsto X_{f,g}$ is constant on chambers and changes by toric birational modifications across walls (proposition 6.6); no vanishing cycle or Dehn twist has been identified. Conjecture 9.2 restates the question.
5. *The m_3 conjecture.* Gluing is strictly associative (theorem 2.3(i)), so each hom-category, regarded as an A_∞ -category, has $m_k = 0$ for $k \geq 3$. The whiskering defect compares $u \cdot (\beta \cdot \alpha)$ with $(u \cdot \beta) \cdot (u \cdot \alpha)$. Its inputs are two 2-cells and a 1-cell, and it measures the failure of whiskering to be a functor. In A_∞ language that is the second component $\mathcal{F}_2$ of an A_∞ -functor, not a ternary composition (now theorem 5.2).
6. *Homotopy category.* The graded shadow has hom-sets $[d(f, g), \infty)$ under addition. These are not the degree-zero cohomology of cochain complexes, so the shadow is not a homotopy category of an A_∞ -category without further structure.
7. *Fisher versus Fisher–Rao.* These are the same metric (proposition 6.1).
8. *References.* The companion tower is now version 9; its posted earlier version has the title “... and Weak Distributive Laws”, a claim withdrawn in the revisions.

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